the world, and he does not block the development of all new medications. In contrast, if all doctors are really just a single system, and that system makes a mistake, the results might be catastrophic. In truth, however, an integrated computer system can maximise the advantages of connectivity without losing the benefits of individuality. You can run many alternative algorithms on the same network, so that a patient in a remote jungle village can access through her smartphone not just a single authoritative doctor, but actually a hundred different AI doctors, whose relative performance is constantly being compared. You don't like what the IBM doctor told you? No problem. Even if you are stranded somewhere on the slopes of Kilimanjaro, you can easily contact the Baidu doctor for a second opinion.

The benefits for human society are likely to be immense. AI doctors could provide far better and cheaper healthcare for billions of people, particularly for those who currently receive no healthcare at all. Thanks to learning algorithms and biometric sensors, a poor villager in an underdeveloped country might come to enjoy far better healthcare via her smartphone than the richest person in the world gets today from the most advanced urban hospital.[5]

Similarly, self-driving vehicles could provide people with much better transport services, and in particular reduce mortality from traffic accidents. Today close to 1.25 million people are killed annually in traffic accidents (twice the number killed by war, crime and terrorism combined).[6] More than 90 per cent of these accidents are caused by very human errors: somebody drinking alcohol and driving, somebody texting a message while driving, somebody falling asleep at the wheel, somebody daydreaming instead of paying attention to the road. The US National Highway Traffic Safety Administration estimated in 2012 that 31 per cent of fatal crashes in the USA involved alcohol abuse, 30 per cent involved speeding, and 21 per cent involved distracted drivers.[7] Self-driving vehicles will never do any of these things. Though they suffer from their own problems and limitations, and though some accidents are inevitable, replacing all human drivers by computers is expected to reduce deaths and injuries on the road by about 90 per cent.[8] In other words, switching to autonomous vehicles is likely to save the lives of a million people every year.

Hence it would be madness to block automation in fields such as transport and healthcare just in order to protect human jobs. After all, what we ultimately ought to protect is humans – not jobs. Redundant drivers and doctors will just have to find something else to do.

The Mozart in the machine